Probability and Statistics

Chapter 8. Fundamental Sampling Distributions and Data Descriptions

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Chapter Overview

8.1 Random Sampling

8.1 Random Sampling

- The outcome of a statistical experiment may be numerical or descriptive.
- Numerical example: total score when two dice are tossed.
- Descriptive example: blood type such as A, B, AB, or O with Rh sign.
- In this chapter, we study sampling from populations and the sampling behavior of statistics.

이 장의 핵심 목적은 sample mean, sample variance와 같은 통계량이 반복 표본추출에서 어떻게 변하는지를 이해하는 것이다.

Why sampling distributions matter

- We usually cannot observe the entire population.
- We observe a sample and compute statistics from that sample.
- Statistical inference uses sample statistics to make conclusions about population parameters.

Population parameter는 고정된 값이지만, statistic은 sample에 따라 달라지는 random variable이다.

Population

Definition 8.1
A **population** consists of the totality of the observations with which we are concerned.

- The population size may be finite or infinite.
- Finite examples: cards in a deck, students in a school, fish in a lake.
- Infinite examples: all future measurements of pressure, all possible battery lifetimes.

Population은 사람만을 의미하지 않는다. 관심 있는 모든 관측값의 전체 집합을 의미한다.

Population as a probability distribution

Each observation in a population can be regarded as a value of a random variable X.

If a population is described by a distribution $f(x)$, then the population mean and variance are the mean and variance of the corresponding random variable.

- Binomial population: observations follow a binomial distribution.
- Normal population: observations follow a normal distribution.
- General population: observations follow some distribution $f(x)$.

Example: Bernoulli population

조립 라인에서 생산된 제품들이 불량 여부에 따라 검사된다고 하자.

- $X = 0$: nondefective item
- $X = 1$: defective item

$$P(X = x) = p^x q^{1-x}, \qquad x = 0, 1,$$

여기서 $q = 1 - p$이다.

이 경우 population은 Bernoulli random variable로 표현된다.

Sample

Definition 8.2
A **sample** is a subset of a population.

- Samples are used because observing the entire population may be impossible or too costly.
- A sample should be representative of the population.
- A biased sampling procedure may consistently overestimate or underestimate a population characteristic.

Random sample

Definition 8.3
Let $X_1, X_2, \ldots, X_n$ be n independent random variables, each having the same probability distribution $f(x)$. Then $X_1, X_2, \ldots, X_n$ is a **random sample** of size n from the population $f(x)$.

$$f(x_1, x_2, \ldots, x_n) = f(x_1)f(x_2)\cdots f(x_n).$$

Random sample의 핵심 조건은 **independent**와 **identically distributed**이다.

Random sample: interpretation

If $n = 8$ batteries are selected from a stable manufacturing process and their lifetimes are recorded,

$$X_1, X_2, \ldots, X_8$$

represent the random sample.

If the population of battery lives is normal, then each X_i has the same normal distribution as the original population.

- The observed data are $x_1, x_2, \ldots, x_8$.
- The random variables before observation are $X_1, X_2, \ldots, X_8$.

8.2 Some Important Statistics

8.2 Some Important Statistics

Our main purpose in random sampling is to obtain information about unknown population parameters.

- Population proportion p may be estimated by sample proportion $\hat{p}$.
- Population mean μ may be estimated by sample mean $\bar{x}$.
- Population variance σ^2 may be estimated by sample variance s^2.

Statistic(통계량)은 관찰값이 주어지기 전에는 random variable(확률변수)이다.

Statistic

Definition 8.4
Any function of the random variables constituting a random sample is called a **statistic**.

Examples:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Statistic(통계량) 은 sample(표본)의 함수이자 **random variable(확률변수)**이다.
Statistic은 sample로부터 계산되므로 sample이 바뀌면 값도 달라진다.

Location measures of a sample

The most common location measures are:

- sample mean
- sample median
- sample mode

Location measure는 데이터의 중심을 나타내지만, 데이터의 spread를 설명하지는 못한다.

Sample mean

Sample mean

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For observed values $x_1, x_2, \ldots, x_n$,

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

- $\bar{X}$ is a statistic.
- $\bar{x}$ is the computed value from observed data.

Sample median

For ordered observations $x_1 \leq x_2 \leq \cdots \leq x_n$,

$$\tilde{x} = \begin{cases} x_{(n+1)/2}, & n \text{ is odd}, \\ \dfrac{x_{n/2} + x_{n/2+1}}{2}, & n \text{ is even}. \end{cases}$$

Median은 extreme value의 영향을 mean보다 훨씬 적게 받는다.

Sample mode

The **sample mode** is the value of the sample that occurs most often.

- A sample can have one mode, more than one mode, or no repeated mode.
- Mode is useful for identifying the most frequent observation.

Example 8.1

다음 관측값들로 이루어진 데이터가 있다고 하자:

$$0.32,\ 0.53,\ 0.28,\ 0.37,\ 0.47,\ 0.43,\ 0.36,\ 0.42,\ 0.38,\ 0.43.$$

sample mode를 구하시오.

이 데이터에서 가장 자주 나타나는 값은 0.43이다. 따라서 sample mode는

$$0.43$$

이다.

Why variability is also needed

Two samples may have the same center but very different variability.

Sample A	0.97	1.00	0.94	1.03	1.06
Sample B	1.06	1.01	0.88	0.91	1.14

Both samples have mean 1.00 liter.

하지만 Sample A가 더 균일하다. 따라서 central tendency만으로는 데이터의 성격을 충분히 설명할 수 없다.

Variability measures of a sample

Important variability measures include:

- sample variance
- sample standard deviation
- sample range

Variability는 observations가 average 주변에서 얼마나 퍼져 있는지를 나타낸다.

Sample variance

Sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

For observed data,

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

분모가 n이 아니라 $n-1$인 이유는 later inference에서 중요하게 다루어진다.

Example 8.2

무작위로 선택된 4개의 식료품점에서 커피 가격을 비교한 결과, 1파운드 봉지 가격이 지난달보다 다음만큼 증가하였다:

$$12,\ 15,\ 17,\ 20$$

센트. 이 표본의 sample variance를 구하시오.

Example 8.2: Solution

먼저 sample mean을 계산하면:

$$\bar{x} = \frac{12 + 15 + 17 + 20}{4} = 16.$$

$$\begin{aligned} s^2 &= \frac{1}{3}\sum_{i=1}^{4}(x_i - 16)^2 \\ &= \frac{(12-16)^2 + (15-16)^2 + (17-16)^2 + (20-16)^2}{3} \\ &= \frac{16+1+1+16}{3} = \frac{34}{3}. \end{aligned}$$

Theorem 8.1

If S^2 is the variance of a random sample of size n, then

$$S^2 = \frac{1}{n(n-1)}\left[n\sum_{i=1}^{n} X_i^2 - \left(\sum_{i=1}^{n} X_i\right)^2\right].$$

이 공식은 계산용 공식으로 유용하다. 특히 $\sum x_i$와 $\sum x_i^2$가 주어졌을 때 편리하다.

Theorem 8.1: Proof (1)

By definition,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Expanding the square,

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}\left(X_i^2 - 2X_i\bar{X} + \bar{X}^2\right).$$

Thus,

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n}X_i^2 - 2\bar{X}\sum_{i=1}^{n}X_i + n\bar{X}^2.$$

Theorem 8.1: Proof (2)

Since

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

we have

$$2\bar{X}\sum_{i=1}^{n} X_i = \frac{2}{n}\left(\sum_{i=1}^{n} X_i\right)^2, \qquad n\bar{X}^2 = \frac{1}{n}\left(\sum_{i=1}^{n} X_i\right)^2.$$

Therefore,

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 = \sum_{i=1}^{n} X_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} X_i\right)^2.$$

Hence,

$$\begin{aligned} S^2 &= \frac{1}{n-1}\left[\sum_{i=1}^{n} X_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} X_i\right)^2\right] \\ &= \frac{1}{n(n-1)}\left[n\sum_{i=1}^{n} X_i^2 - \left(\sum_{i=1}^{n} X_i\right)^2\right]. \end{aligned}$$

Sample standard deviation and sample range

Sample standard deviation

$$S = \sqrt{S^2}.$$

Sample range

$$R = X_{\text{max}} - X_{\text{min}}.$$

- Standard deviation has the same unit as the data.
- Range is easy to compute but strongly affected by extreme values.

Example 8.3

다음 데이터의 variance를 구하시오:

$$3,\ 4,\ 5,\ 6,\ 6,\ 7,$$

이는 무작위로 선택된 6명의 낚시꾼이 잡은 송어의 수를 나타낸다.

Example 8.3: Solution

다음 값을 얻는다:

$$\sum_{i=1}^{6} x_i = 31, \qquad \sum_{i=1}^{6} x_i^2 = 171, \qquad n = 6.$$

Theorem 8.1을 사용하면,

$$\boxed{\begin{aligned} s^2 &= \frac{1}{(6)(5)}\left[(6)(171) - (31)^2\right] \\ &= \frac{1026 - 961}{30} = \frac{65}{30} = \frac{13}{6}. \end{aligned}}$$

따라서,

$$s = \sqrt{\frac{13}{6}} \approx 1.47, \qquad R = 7 - 3 = 4.$$

8.3 Sampling Distributions

8.3 Sampling Distributions

Statistical inference is concerned with generalizations and predictions based on samples.

- We compute a statistic from a sample.
- We use the statistic to make statements about population parameters.
- Because a statistic varies from sample to sample, it has a probability distribution.

Sampling distribution

Definition 8.5
The probability distribution of a statistic is called a **sampling distribution**.

Sampling distribution은 한 번 얻은 sample이 아니라, 같은 크기의 sample을 반복해서 뽑았을 때 statistic이 어떻게 분포하는지를 설명한다.

Sampling distribution of the mean

The probability distribution of $\bar{X}$ is called the **sampling distribution of the mean**.

- It describes the variability of sample means around μ.
- It tells us how much an observed $\bar{x}$ can differ from the true mean by chance.

Sampling distribution of S^2

The sampling distribution of S^2 describes the variability of sample variances around σ^2.

For inference on population variability, the statistic S^2 is the natural counterpart to the parameter σ^2.

- $\bar{X}$ is used for inference on μ.
- S^2 is used for inference on σ^2.

8.4 Sampling Distribution of Means and the Central Limit Theorem

8.4 Sampling Distribution of Means

Suppose a random sample of size n is taken from a normal population with mean μ and variance σ^2.

$$\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n).$$

Then $\bar{X}$ is normally distributed with

$$\mu_{\bar{X}} = \mu, \qquad \sigma^2_{\bar{X}} = \frac{\sigma^2}{n}.$$

Standard error of the sample mean

The standard deviation of $\bar{X}$ is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

This is often called the **standard error** of the sample mean.

표본크기 n이 커질수록 sample mean의 변동성은 감소한다.

Central Limit Theorem

Theorem 8.2: Central Limit Theorem

If $\bar{X}$ is the mean of a random sample of size n taken from a population with mean μ and finite variance σ^2, then the limiting form of the distribution of

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

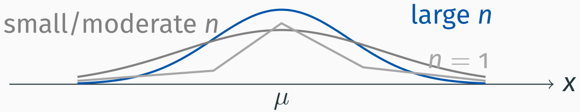
as $n \to \infty$ is the standard normal distribution $N(0, 1)$.

Central Limit Theorem: interpretation

- If the original population is normal, $\bar{X}$ is exactly normal for any n.
- If the population distribution is unknown, $\bar{X}$ is approximately normal for large n.
- A common guideline is $n \geq 30$, unless the population is strongly skewed.

Central Limit Theorem은 원자료의 분포가 정규분포가 아니어도 sample mean의 분포가 정규분포에 가까워진다는 매우 중요한 결과이다.

Illustration of the Central Limit Theorem



n이 커질수록 $\bar{X}$의 분포는 더 정규분포에 가까워지고, 분산은 σ^2/n으로 작아진다.

Example 8.4

한 전기 회사가 생산하는 전구의 수명은 다음 평균과 표준편차를 갖는 근사적인 normal distribution을 따른다고 하자:

$$\mu = 800 \text{ hours}, \qquad \sigma = 40 \text{ hours}.$$

무작위로 선택한 16개 전구의 평균 수명이 775시간보다 작을 확률을 구하시오.

전구 수명 평균이 800시간이고 표준편차가 40시간일 때, 16개 표본의 평균 수명이 775시간보다 작을 확률을 구한다.

Example 8.4: Solution

$n = 16$이므로,

$$\mu_{\bar{X}} = 800, \qquad \sigma_{\bar{X}} = \frac{40}{\sqrt{16}} = 10.$$

$$z = \frac{775 - 800}{10} = -2.5.$$

따라서,

$$P(\bar{X} < 775) = P(Z < -2.5) = 0.0062.$$

Inferences on the population mean

The Central Limit Theorem can be used to judge whether an observed sample mean is reasonable under a proposed population mean.

If an observed $\bar{x}$ is extremely unlikely when a claimed μ is true, then the data cast doubt on that claimed value of μ.

- This idea leads to hypothesis testing.
- Formal procedures are introduced in later chapters.

Case Study 8.1: Automobile Parts

한 제조공정에서 원통형 부품을 생산한다. 엔지니어는 population mean diameter가 다음과 같다고 추측한다:

$$\mu = 5.0 \text{ mm}.$$

무작위로 선택된 100개 부품에서 다음 결과가 얻어졌다:

$$\bar{x} = 5.027 \text{ mm}, \qquad \sigma = 0.1 \text{ mm}.$$

이 표본 정보는 엔지니어의 추측을 지지하는가, 아니면 반박하는가?

표본평균이 5.027mm가 나온 것이 $\mu = 5.0$이라는 가정 아래에서 얼마나 드문 사건인지 확인한다.

Case Study 8.1: Solution (1)

$\mu = 5.0$이 참이라면,

$$\sigma_{\bar{X}} = \frac{0.1}{\sqrt{100}} = 0.01.$$

다음을 계산한다:

$$P(|\bar{X} - 5| \geq 0.027).$$

$$\boxed{P(|\bar{X} - 5| \geq 0.027) = 2P\left(\frac{\bar{X} - 5}{0.1/\sqrt{100}} \geq 2.7\right).}$$

Case Study 8.1: Solution (2)

$$2P(Z \geq 2.7) = 2(0.0035) = 0.007.$$

$\mu = 5.0$이 참이라면, 5.0에서 적어도 0.027 mm 이상 떨어진 sample mean을 관측하는 일은 1000번의 실험 중 약 7번 정도만 발생한다.

따라서 $\bar{x} = 5.027$은 $\mu = 5.0$이라는 conjecture를 강하게 의심하게 만드는 결과이다.

Example 8.5

대학의 두 캠퍼스 사이를 셔틀버스로 이동하는 데 걸리는 평균 시간은

$$\mu = 28 \text{ minutes}, \qquad \sigma = 5 \text{ minutes}.$$

어느 한 주 동안 버스는 승객을 40번 운송하였다. 평균 운송 시간이 30분보다 클 확률을 구하시오. 평균 시간은 가장 가까운 분 단위로 측정된다고 가정한다.

40번 운행의 평균 시간이 30분보다 클 확률을 구한다.

Example 8.5: Solution

평균 시간이 가장 가까운 분 단위로 측정되므로,

$$\bar{X} > 30 \quad \text{은} \quad \bar{X} \geq 30.5 \quad \text{로 처리된다.}$$

$$P(\bar{X} > 30) = P\left(\frac{\bar{X} - 28}{5/\sqrt{40}} \geq \frac{30.5 - 28}{5/\sqrt{40}}\right)$$
$$= P(Z \geq 3.16) = 0.0008.$$

Difference between two sample means

Suppose two independent samples are drawn from two populations:

$$(\mu_1, \sigma_1^2), \qquad (\mu_2, \sigma_2^2).$$

The statistic of interest is

$$\bar{X}_1 - \bar{X}_2.$$

It is used to compare two population means.

Mean and variance of $\bar{X}_1 - \bar{X}_2$

For independent samples,

$$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2,$$

$$\sigma^2_{\bar{X}_1 - \bar{X}_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

두 sample이 independent이므로 variance가 더해진다.

Theorem 8.3

If independent samples of sizes n_1 and n_2 are drawn from two populations with means μ_1, μ_2 and variances σ_1^2, σ_2^2, then

$$\bar{X}_1 - \bar{X}_2$$

is approximately normally distributed with

$$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2, \qquad \sigma^2_{\bar{X}_1 - \bar{X}_2} = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}.$$

Theorem 8.3: standardized form

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{(\sigma_1^2/n_1) + (\sigma_2^2/n_2)}}$$

is approximately standard normal.

- The approximation is good when n_1, n_2 are large.
- If both populations are normal, the result is exact for any sample sizes.

Case Study 8.2: Paint Drying Time

두 종류의 페인트 A와 B를 비교한다.

- $n_A = n_B = 18$
- $\sigma_A = \sigma_B = 1.0$
- $\mu_A = \mu_B$라고 가정한다

다음을 구하시오:

$$P(\bar{X}_A - \bar{X}_B > 1.0).$$

두 페인트의 평균 건조 시간이 같다고 가정할 때, 표본평균 차이가 1시간보다 클 확률을 구한다.

Case Study 8.2: Solution

$\mu_A = \mu_B$이면,

$$\mu_{\bar{X}_A - \bar{X}_B} = 0.$$

또한,

$$\sigma^2_{\bar{X}_A - \bar{X}_B} = \frac{1}{18} + \frac{1}{18} = \frac{1}{9}.$$

$$z = \frac{1.0 - 0}{\sqrt{1/9}} = 3.0.$$

$$P(\bar{X}_A - \bar{X}_B > 1.0) = P(Z > 3.0) = 0.0013.$$

What do we learn from Case Study 8.2?

- $\mu_A = \mu_B$라면 1시간의 차이는 매우 드문 결과이다.
- 따라서 이러한 결과는 두 population mean이 같지 않을 수 있다는 증거가 된다.

그러나 관측된 차이가 0.25시간에 불과하다면,

$$P(\bar{X}_A - \bar{X}_B > 0.25) = P(Z > 0.75) = 0.2266.$$

이 확률은 작지 않으므로, population mean이 다르다는 강한 증거로 보기는 어렵다.

Example 8.6

제조업체 A의 텔레비전 picture tube 수명은

$$\mu_1 = 6.5, \quad \sigma_1 = 0.9,$$

제조업체 B의 경우는

$$\mu_2 = 6.0, \quad \sigma_2 = 0.8.$$

A에서 무작위로 선택한 36개 tube의 평균 수명이 B에서 선택한 49개 tube의 평균 수명보다 적어도 1년 이상 길 확률을 구하시오.

Example 8.6: Solution (1)

다음 값을 얻는다:

$$n_1 = 36, \qquad n_2 = 49.$$

$\bar{X}_1 - \bar{X}_2$의 평균은

$$\mu_{\bar{X}_1 - \bar{X}_2} = 6.5 - 6.0 = 0.5.$$

표준편차는

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{0.9^2}{36} + \frac{0.8^2}{49}} = \sqrt{\frac{0.81}{36} + \frac{0.64}{49}} \approx 0.189.$$

Example 8.6: Solution (2)

구해야 할 확률은

$$P(\bar{X}_1 - \bar{X}_2 \geq 1.0).$$

$$z = \frac{1.0 - 0.5}{0.189} = 2.65.$$

$$P(\bar{X}_1 - \bar{X}_2 \geq 1.0) = P(Z > 2.65) = 1 - 0.9960 = 0.0040.$$

Normal approximation to the binomial revisited

The Central Limit Theorem also explains why the binomial distribution can be approximated by a normal distribution.

A binomial random variable X is the sum of n independent Bernoulli random variables.

$$X = X_1 + X_2 + \cdots + X_n, \qquad X_i \in \{0, 1\}.$$

The sample proportion X/n is an average of 0s and 1s. Therefore, for large n, X/n and X are approximately normal under suitable conditions.

Condition for normal approximation to binomial

From Chapter 6, the usual guideline is

$$np \geq 5, \qquad nq \geq 5,$$

where $q = 1 - p$.

이 조건은 Bernoulli observations의 평균이 정규분포에 가까워질 만큼 success와 failure가 충분히 관측되는지를 확인하는 기준이다.

8.5 Sampling Distribution of S^2

8.5 Sampling Distribution of S^2

The sampling distribution of S^2 is used for inference about the population variance σ^2.

If the population is normal, then a scaled version of the sample variance follows a chi-squared distribution.

Decomposition of total variation

If a random sample of size n is drawn from a normal population with mean μ and variance σ^2, then

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}[(X_i - \bar{X}) + (\bar{X} - \mu)]^2.$$

This becomes

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \mu)^2.$$

Dividing by σ^2,

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 = \frac{(n-1)S^2}{\sigma^2} + \frac{(\bar{X} - \mu)^2}{\sigma^2/n}.$$

Theorem 8.4

If S^2 is the variance of a random sample of size n taken from a normal population having variance σ^2, then

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\sigma^2}$$

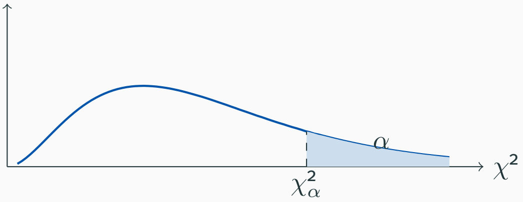
has a chi-squared distribution with

$$v = n - 1$$

degrees of freedom.

Chi-squared critical values

It is customary to let χ^2_α represent the chi-squared value above which we find an area of α.



Chi-squared distribution은 symmetric하지 않으므로 오른쪽 tail과 왼쪽 tail 값을 따로 확인해야 한다.

Example 8.7

자동차 배터리 제조업체는 배터리 수명이 평균 3년, 표준편차 1년이라고 보증한다. 5개의 배터리 수명은 다음과 같다:

$$1.9,\ 2.4,\ 3.0,\ 3.5,\ 4.2$$

년. 배터리 수명이 normal distribution을 따른다고 가정할 때, 제조업체는 여전히 표준편차가 1년이라고 확신해도 되는가?

Example 8.7: Solution (1)

Theorem 8.1을 사용하면,

$$s^2 = \frac{(5)(48.26) - (15)^2}{(5)(4)} = 0.815.$$

$\sigma^2 = 1$이라면,

$$\boxed{\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{4(0.815)}{1} = 3.26.}$$

이는 다음 degrees of freedom을 갖는 chi-squared distribution을 따른다:

$$v = n - 1 = 4$$

degrees of freedom.

Example 8.7: Solution (2)

$v = 4$일 때, chi-squared 값의 95%는 다음 구간에 놓인다:

$$0.484 \quad \text{and} \quad 11.143.$$

계산된 값

$$\chi^2 = 3.26$$

은 이 구간 안에 있다.

따라서 $\sigma = 1$이라는 주장은 이 표본으로는 의심할 만한 충분한 근거가 없다.

Degrees of freedom

When μ is unknown and replaced by $\bar{X}$, one degree of freedom is lost.

- If μ is known:

$$\sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2_n.$$

- If μ is estimated by $\bar{X}$:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

8.6 t-Distribution

8.6 t-Distribution

In the Central Limit Theorem, we used

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

But in practice, σ is often unknown.

When σ is unknown, we replace it by the sample standard deviation S and use

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

Why the t-distribution is needed

- S varies from sample to sample.
- For small samples, this extra variability is not negligible.
- Therefore, T does not follow the standard normal distribution exactly.

If the original population is normal, then T follows a t-distribution with $n - 1$ degrees of freedom.

Relationship with Z and chi-squared

For a normal random sample,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1),$$

and

$$V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Then

$$T = \frac{Z}{\sqrt{V/(n-1)}}.$$

Theorem 8.5

Let Z be a standard normal random variable and V a chi-squared random variable with v degrees of freedom. If Z and V are independent, then

$$T = \frac{Z}{\sqrt{V/v}}$$

has the **t-distribution** with v degrees of freedom.

For completeness, the density is

$$h(t) = \frac{\Gamma[(v+1)/2]}{\Gamma(v/2)\sqrt{\pi v}} \left(1 + \frac{t^2}{v}\right)^{-(v+1)/2}, \qquad -\infty < t < \infty.$$

Corollary 8.1

Let $X_1, X_2, \ldots, X_n$ be independent normal random variables with mean μ and standard deviation σ. Let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a t-distribution with

$$v = n - 1$$

degrees of freedom.

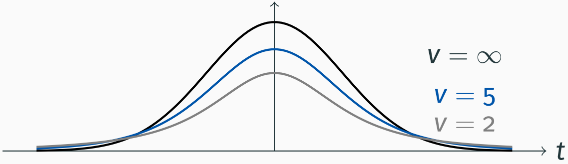
Student t-distribution

The t-distribution was first published by W. S. Gosset in 1908 under the name “Student.”

The t-distribution is often called the **Student t-distribution**.

- It is symmetric about 0.
- It is bell-shaped.
- It has heavier tails than the standard normal distribution.
- As $v \to \infty$, the t-distribution approaches the standard normal distribution.

Shape of the t-distribution



Degrees of freedom이 작을수록 tail이 두껍고, 표본에서 오는 불확실성이 더 크게 반영된다.

t critical values

It is customary to let t_α represent the t-value above which we find an area of α.

Because the t-distribution is symmetric about 0,

$$t_{1-\alpha} = -t_\alpha.$$

For example,

$$t_{0.95} = -t_{0.05}, \qquad t_{0.99} = -t_{0.01}.$$

Example 8.8

$v = 14$ degrees of freedom을 갖는 t-distribution에서 왼쪽 면적이 0.025가 되게 하는 t-value를 구하시오.

왼쪽 면적이 0.025라는 것은 오른쪽 면적이 0.975라는 뜻이므로,

$$t_{0.975} = -t_{0.025}.$$

$v = 14$인 t-table에서,

$$t_{0.025} = 2.145.$$

따라서,

$$t_{0.975} = -2.145.$$

Example 8.9

다음을 구하시오:

$$P(-t_{0.025} < T < t_{0.05}).$$

$t_{0.05}$는 오른쪽 꼬리 면적을 0.05 남기고, $-t_{0.025}$는 왼쪽 꼬리 면적을 0.025 남기므로,

$$P(-t_{0.025} < T < t_{0.05}) = 1 - 0.025 - 0.05 = 0.925.$$

Example 8.10

다음을 만족하는 k를 구하시오:

$$P(k < T < -1.761) = 0.045$$

normal distribution에서 선택된 크기 15의 random sample에 대해 구하시오.

여기서,

$$v = n - 1 = 14.$$

t-table에서 $v = 14$일 때 $1.761 = t_{0.05}$이므로,

$$-1.761 = -t_{0.05}.$$

Example 8.10: Solution

다음과 같이 두자:

$$k = -t_{\alpha}.$$

k와 $-t_{0.05}$ 사이의 면적은 0.045이다.

$$0.045 = 0.05 - \alpha \quad \Rightarrow \quad \alpha = 0.005.$$

따라서,

$$k = -t_{0.005} = -2.977.$$

따라서,

$$P(-2.977 < T < -1.761) = 0.045.$$

Example 8.11

한 화학공학자는 어떤 batch process의 population mean yield가 다음과 같다고 주장한다:

$$\mu = 500$$

grams per milliliter of raw material이라고 한다. 그는 매달 25개의 batch를 표본으로 선택한다. 계산된 t-value가 $-t_{0.05}$와 $t_{0.05}$ 사이에 있으면 이 주장을 받아들인다.

한 표본에서,

$$\bar{x} = 518, \qquad s = 40.$$

분포가 근사적으로 normal이라고 가정할 때, 어떤 결론을 내려야 하는가?

Example 8.11: Solution

여기서,

$$n = 25, \qquad v = 24.$$

t-table에서,

$$t_{0.05} = 1.711.$$

$\mu = 500$이라면,

$$\boxed{t = \frac{518 - 500}{40/\sqrt{25}} = \frac{18}{8} = 2.25.}$$

다음이므로

$$2.25 > 1.711,$$

계산된 t-value는 허용 구간 밖에 있다.

Example 8.11: Interpretation

$v = 24$에서 2.25 이상의 t-value가 나올 확률은 약 0.02이다.

따라서 $\mu = 500$이라는 claim은 이 표본과 잘 맞지 않는다. 표본평균이 더 크므로, process가 engineer가 생각한 것보다 더 높은 yield를 내고 있을 가능성이 있다.

Use of the t-distribution

The t-distribution is used extensively for inference about a population mean when σ is unknown.

- one-sample inference on μ
- comparison of two sample means
- confidence intervals and hypothesis tests in later chapters

The statistic $T = (\bar{X} - \mu)/(S/\sqrt{n})$ requires normality of the original population for the exact t-distribution result.

8.7 F-Distribution

8.7 F-Distribution

The F-distribution is used to compare variances.

The statistic F is defined as the ratio of two independent chi-squared random variables, each divided by its degrees of freedom:

$$F = \frac{U/v_1}{V/v_2}.$$

- $U \sim \chi^2_{v_1}$
- $V \sim \chi^2_{v_2}$
- U and V are independent

Theorem 8.6

Let U and V be independent chi-squared random variables with v_1 and v_2 degrees of freedom. Then

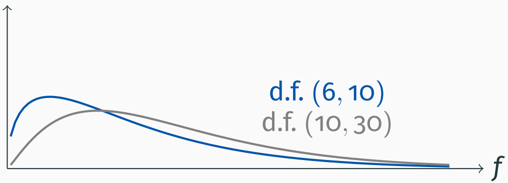
$$F = \frac{U/v_1}{V/v_2}$$

has an F-distribution with v_1 and v_2 degrees of freedom.

For completeness, the density is

$$h(f) = \frac{\Gamma[(v_1+v_2)/2](v_1/v_2)^{v_1/2}}{\Gamma(v_1/2)\Gamma(v_2/2)} \frac{f^{(v_1/2)-1}}{(1+v_1 f/v_2)^{(v_1+v_2)/2}}, \quad f > 0.$$

Shape of the F-distribution



F-distribution depends on two degrees of freedom, and the order (v_1, v_2) matters.

F critical values

Let f_α be the f-value above which we find an area equal to α.

For example, if $v_1 = 6$ and $v_2 = 10$, the f-value leaving area 0.05 to the right is

$$f_{0.05} = 3.22.$$

F-distribution is not symmetric, so left-tail values are obtained using a reciprocal relationship.

Theorem 8.7

Writing $f_\alpha(v_1, v_2)$ for the F critical value with v_1 and v_2 degrees of freedom,

$$f_{1-\alpha}(v_1, v_2) = \frac{1}{f_\alpha(v_2, v_1)}.$$

Example:

$$f_{0.95}(6, 10) = \frac{1}{f_{0.05}(10, 6)} = \frac{1}{4.06} = 0.246.$$

F-distribution with two sample variances

Suppose independent random samples of sizes n_1 and n_2 are selected from two normal populations with variances σ_1^2 and σ_2^2.

From the chi-squared result,

$$\chi_1^2 = \frac{(n_1 - 1)S_1^2}{\sigma_1^2}, \qquad \chi_2^2 = \frac{(n_2 - 1)S_2^2}{\sigma_2^2}.$$

Theorem 8.8

If S_1^2 and S_2^2 are the variances of independent random samples of sizes n_1 and n_2 taken from normal populations with variances σ_1^2 and σ_2^2, then

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}$$

has an F-distribution with

$$v_1 = n_1 - 1, \qquad v_2 = n_2 - 1.$$

What is the F-distribution used for?

- comparing two population variances
- variance ratio methods
- analysis of variance
- comparing variability within samples and between samples

F-distribution은 variance ratio distribution이라고도 불린다.

Motivation for analysis of variance

Suppose three paints A, B, and C are compared.

Paint	Sample Mean	Sample Variance	Sample Size
A	$\bar{X}_A = 4.5$	$s_A^2 = 0.20$	10
B	$\bar{X}_B = 5.5$	$s_B^2 = 0.14$	10
C	$\bar{X}_C = 6.5$	$s_C^2 = 0.11$	10

We want to determine whether the population means are equivalent.

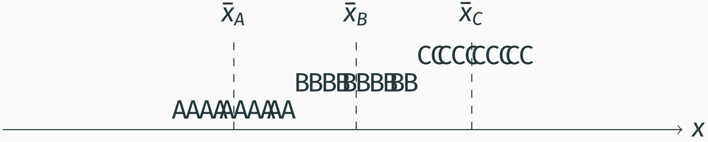
Two sources of variability

In analysis of variance, two types of variability are compared.

1. Variability within samples
2. Variability between sample averages

- If within-sample variability is large, sample overlap can be large even when sample means differ.
- If between-sample variability is large relative to within-sample variability, the population means may differ.

Graphical idea for comparing samples



Sample means are far apart relative to within-sample spread, suggesting possible differences among population means.

8.8 Quantile and Probability Plots

8.8 Quantile and Probability Plots

Graphical tools help us understand the shape of sample data.

- stem-and-leaf plots
- histograms
- box-and-whisker plots
- quantile plots
- normal quantile-quantile plots

These plots help diagnose whether assumptions such as normality are reasonable.

Quantile

Definition 8.6
A **quantile** of a sample, $q(f)$, is a value for which a specified fraction f of the data values is less than or equal to $q(f)$.

Examples:

- median: $q(0.5)$
- lower quartile: $q(0.25)$
- upper quartile: $q(0.75)$

Quantile plot

Let ordered observations be

$$y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(n)}.$$

The quantile plot graphs $y_{(i)}$ against

$$f_i = \frac{i - 3/8}{n + 1/4}.$$

Quantile plot은 모든 observations를 보여주며, median과 quartiles를 시각적으로 확인할 수 있게 해준다.

Interpreting a quantile plot

- Flat regions indicate many observations clustered near the same value.
- Steep regions indicate sparse observations.
- The plot gives a visual approximation to the cumulative distribution structure.

Boxplot은 일부 요약값을 보여주지만, quantile plot은 모든 데이터 값을 시각화한다.

Normal quantile-quantile plot

A normal quantile-quantile plot compares ordered observations with corresponding theoretical normal quantiles.

For the standard normal distribution, an approximation to the quantile is

$$q_{0,1}(f) = 4.91 \left[f^{0.14} - (1 - f)^{0.14} \right].$$

Definition 8.7

The **normal quantile-quantile plot** is a plot of $y_{(i)}$ against $q_{0,1}(f_i)$, where

$$f_i = \frac{i - 3/8}{n + 1/4}.$$

A nearly straight-line relationship suggests that the data came from a normal distribution.

Interpreting normal Q-Q plots

- Approximate straight line: normality is plausible.
- Strong curvature: skewness or nonnormality may be present.
- Outlying points: possible extreme observations or heavy tails.

The intercept estimates the population mean μ, and the slope estimates the standard deviation σ.

Normal probability plotting

Normal probability plots are another way to assess normality.

- The ordered data values are plotted against a transformed probability scale.
- A straight-line pattern supports the normality assumption.
- Probability plots can be adapted to distributions other than normal.

그래프는 formal test보다 직관적으로 normality assumption이 합리적인지 보여줄 수 있다.

Example 8.12

하수 스트레스에 대한 영양염류 보존과 대형 무척추동물 군집 반응 연구에서, 두 개의 서로 다른 station에서 밀도 측정값이 수집되었다.

normal quantile-quantile plot을 작성하고, 두 표본이 동일한 $N(\mu, \sigma)$ distribution에서 나왔다고 가정하는 것이 타당한지 판단하는 것이 목표이다.

Example 8.12: Data

Station 1	Station 2
5030	2800
13700	4670
10730	6890
11400	7720
860	7030
2200	7330
4250	2810
15040	1330
4980	3320
11910	1230
8130	2130
26850	2190
17660	
22800	
1130	
1690	

Example 8.12: Interpretation

normal quantile-quantile plot은 하나의 직선 형태와 거리가 멀다.

Station 1의 데이터는 lower tail과 upper tail에 여러 값이 나타나며, 관측값들이 서로 다른 영역에 군집되어 있는 것처럼 보인다.

따라서 두 station의 데이터가 동일한 $N(\mu, \sigma)$ 분포에서 왔다고 보기 어렵다.

8.9 Potential Misconceptions and Hazards

8.9 Potential Misconceptions and Hazards

The Central Limit Theorem is one of the most powerful tools in statistics, but it must be used carefully.

The concept of a sampling distribution is fundamental. Inference from a single observed statistic must be based on the theoretical distribution of all possible statistic values from repeated samples of the same size.

Misconception 1: Using CLT without knowing σ

One cannot directly use

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

unless σ is known.

If σ is unknown, it should be replaced by S, and the resulting statistic may require the t-distribution rather than the standard normal distribution.

Misconception 2: Confusing CLT and t-distribution

The statistic

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

is not a direct result of the Central Limit Theorem.

- For exact t-distribution theory, the sample must come from a normal population.
- S is an estimate of σ.
- The degrees of freedom reflect the amount of sample information.

Misconception 3: Forgetting the source of t, χ^2, and F

The t, χ^2, and F distributions arise naturally from samples drawn from normal populations.

- t: inference about a mean when σ is unknown
- χ^2: inference about a variance
- F: comparison of variances and analysis of variance

Connection to later chapters

Sampling distributions provide the foundation for later topics:

- parameter estimation
- confidence intervals
- hypothesis testing
- analysis of variance
- goodness-of-fit tests

이 장의 내용은 이후 통계적 추론의 모든 계산과 판단의 기초가 된다.